Applications of Neural Networks in Continous and Dynamic Systems

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Introduction

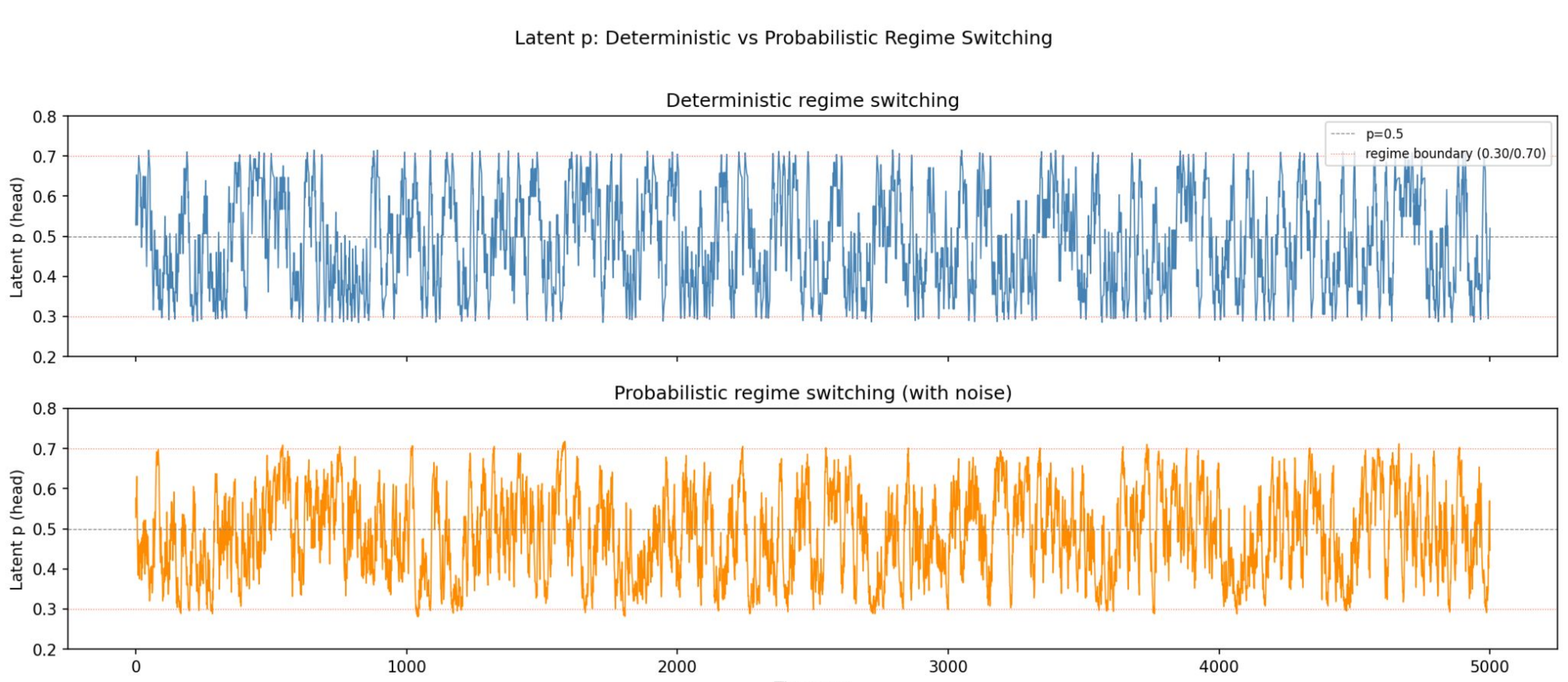
Many real-world processes violate the assumption of stationarity, making them difficult to model with traditional statistical methods. While substantial research since 2015 has focused on developing models for dynamic and temporal systems, most approaches evaluate performance primarily in terms of predictive accuracy.

In contrast, this work examines whether neural network based models can recover the underlying stochastic dynamics of non-stationary systems. Rather than emphasizing prediction, we assess how well each model captures the evolving structural behavior of the data-generating process. We evaluate six architectures, autoregressive MLP, RNN, LSTM, Transformer, Neural ODE, and Neural Jump ODE, on their ability to learn these dynamics.

Methods

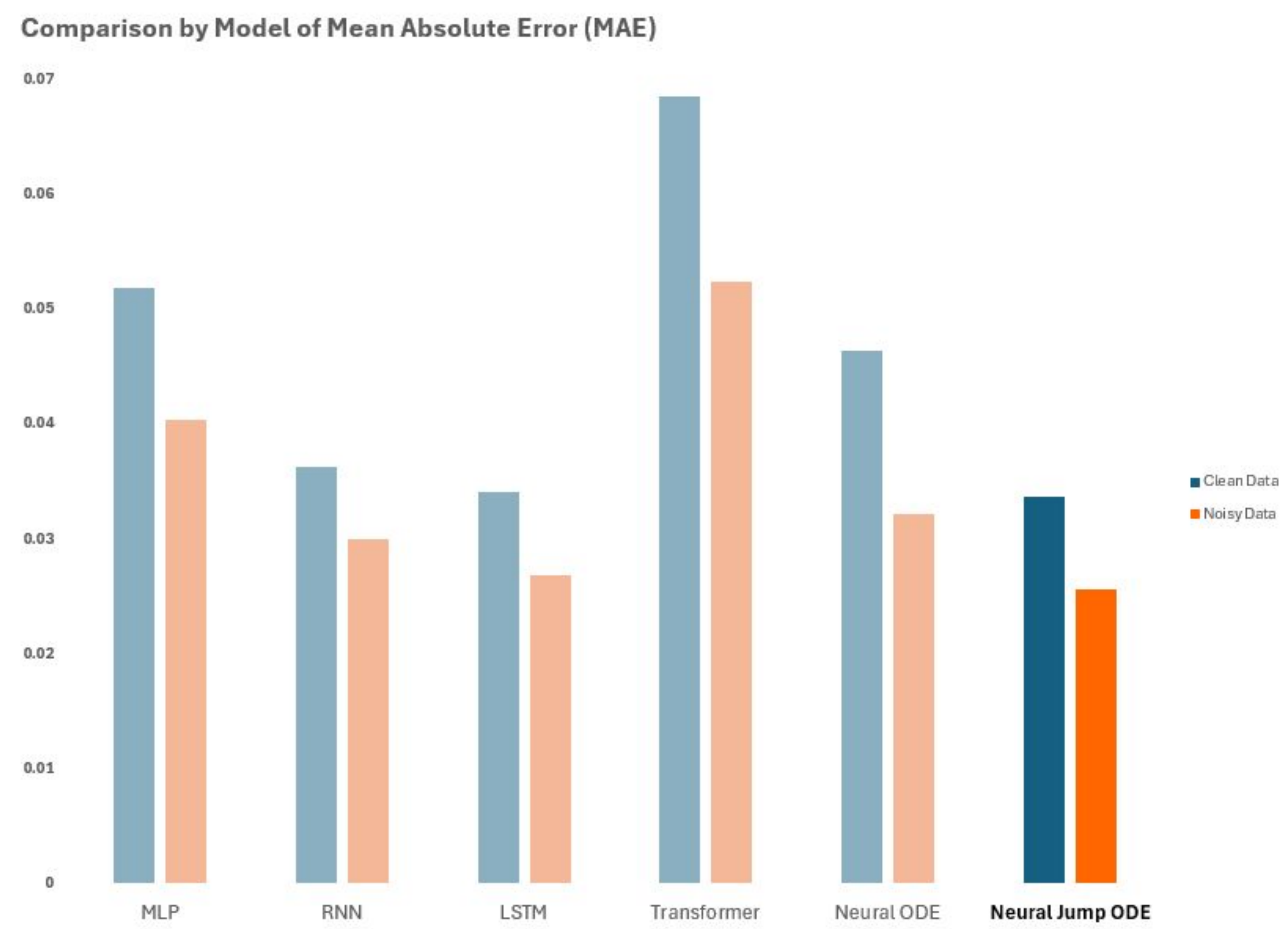
For our synthetic data, we generated two non-stationary Bernoulli processes with regime-switching dynamics. In both cases, the latent probability 'p' evolved as a function of prior outcomes, alternating between upward/downward trending and mean-reverting regimes. The 'clean dataset' followed a deterministic regime-switching mechanism, where transitions occurred at predefined thresholds of p. The 'noisy dataset' employed probabilistic regime transitions and added Gaussian noise to p at each step. The datasets can be seen in the figure below, with the 'clean dataset' on top in blue and the 'noisy dataset' on bottom in orange.

The models were trained using binary cross-entropy loss to predict observed outcomes. However, model evaluation focused on their ability to recover the latent probability of p, using the final sigmoid output as an estimate. Model performance was assessed using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and correlation between predicted and true values of p.



Results

All models were trained on the training sets and evaluated on the test sets. Model performance was compared using the three evaluation criteria; MAE, RMSE, and correlation. The figure below shows the test results for each model for the evaluation metric MAE.

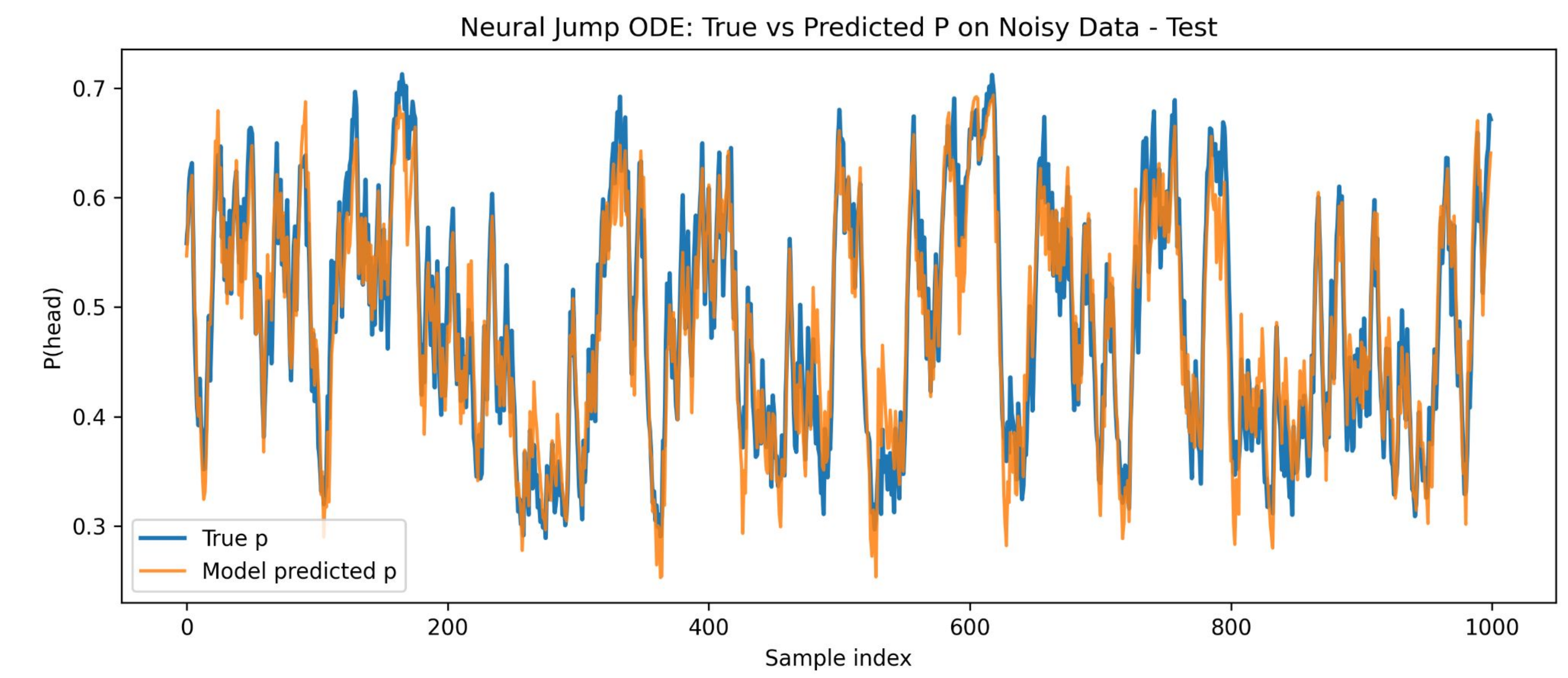


All three evaluation metrics were considered in identifying the best performing model. The table below summarizes the performance of each architecture on both the clean and noisy datasets. The Neural Jump ODE consistently achieved the strongest results across all metrics. Notably, its performance improved on the noisy dataset relative to the clean dataset. The LSTM ranked second and exhibited a similar pattern, achieving improved performance on the noisy dataset relative to the clean dataset.

	Clean Data			Noisy Data		
	MAE	RMSE	Corr	MAE	RMSE	Corr
MLP	0.0518	0.0658	0.8493	0.0404	0.0506	0.8911
RNN	0.0362	0.0472	0.9266	0.03	0.0374	0.9467
LSTM	0.034	0.0444	0.9373	0.0268	0.0335	0.9473
Transformer	0.0685	0.0864	0.7207	0.0524	0.0648	0.7725
Neural ODE	0.0464	0.0601	0.8827	0.0321	0.0401	0.9291
Neural Jump ODE	**0.0336**	**0.0435**	**0.9374**	**0.0256**	**0.0324**	**0.9521**

Best Performing Model

Second Best Performing Model



The figure above illustrates the Neural Jump ODE model's predicted probability p (orange) compared to the true underlying probability (blue). The relatively close alignment of the two lines demonstrates the ability of the model to recover the underlying latent dynamics governing the data-generating process.

Conclusions

Based on our findings, several neural network architectures demonstrate a strong capability in modeling non-stationary stochastic processes, particularly in the presence of regime-switching dynamics. Across all experiments, recurrent architectures (RNN, LSTM) outperform a simple autoregressive MLP baseline. Surprisingly, the transformer model was the worst performing model across all experiments. Based on these results, further experimentation with fine-tuning the transformer architecture may be warranted to see if different results can be achieved.

Continuous-time approaches further improve performance, with Neural ODE models achieving competitive results and the Neural Jump ODE consistently performing best across all evaluation metrics. This suggests that explicitly modeling regime changes and temporal dynamics is critical for accurately capturing non-stationary behavior.

Overall, our results indicate that neural architectures, especially those designed to handle discontinuities, are well-suited for learning complex, time-varying stochastic systems. Future work will extend these methods to real-world datasets and more complex non-stationary environments.

References

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